Course Name: Operating Systems

Course Code: MCA 125

Objectives:

The objective of the course is to enable students to

- Understand the functionalities and internals of operating systems
- Optimize their programs to execute on any operating system
- Able to implement tasks using Linux shell scripting

Prerequisites:

Fundamentals of Computers

Contents:

1. **Overview of Operating Systems**
   Role of Operating Systems (OS), Operations of OS, Resource Management, Security and Protection, Virtualization, Distributed Systems, Kernel Data Structures, Computing Environments, Free and Open-Source OS, OS Services, User and OS Interface, System Calls, System Services, OS Specific applications, OS Design and Implementation, OS Structure, Building and Booting an OS, Overview of Containers and Application Containerization

2. **Process Management**
   Process Concept, Process Scheduling, Operations on Process, Interprocess Communication, IPC in Shared-Memory Systems and Message-Passing Systems, Examples of IPC Systems, Communication in Client-Server Systems, Threads & Concurrency, Multicore Programming, Multithreading Models, Thread Libraries, Implicit Threading, Threading Issues, Concepts of CPU Scheduling, Scheduling Criteria, Scheduling Algorithms, Thread Scheduling, Multiprocessor Scheduling, Overview of Real-Time Scheduling, Algorithm Evaluation

3. **Process Synchronization**
   Overview of Synchronization Tools, The Critical-Section Problem, Peterson's Solution, Hardware Support for Synchronization, Mutex Locks, Semaphores, Monitors, Liveness, Evaluation, Classical Problems of Synchronization, POSIX and Java Synchronization, Overview of Deadlocks, Deadlock in Multithreaded Applications, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock

4. **Memory Management**
   Background, Contiguous Memory Allocation, Paging, Structure of the Page Table, Swapping, Virtual Memory Overview, Demand Paging, Copy-on-Write, Page Replacement, Allocation of Frames, Thrashing, Memory Compression, Allocating Kernel Memory, Considerations for Virtual Memory Management

5. **Storage Management**
   Overview of Mass Storage Structure, HDD Scheduling, NVM Scheduling, Error Detection and Correction, Storage Device Management, Swap-Space Management, Storage Attachment, RAID Structure, Overview of I/O Systems, Application I/O Interface, Streams, Performance of I/O

6. **File System**
   Overview of File-System Interface, File-System Structure, File-System Operations, Directory Implementation, Allocation Methods, Free-Space Management, Efficiency and Performance, Recovery, File Systems, File-System Mounting, Partitions and Mounting, File Sharing, Virtual File Systems, Remote File Systems, NFS

7. **Linux Shell Scripting**
   Basic commands of Linux, Process related commands of Linux, Linux APIs for process management and IPC, File related commands of Linux, Linux Shell Scripting to automate tasks, Processing files using Shell Scripts

References:

1. Silberschatz, A., Peter B. Galvin and Greg Gagne, "*Operating System Concepts*", 10th Edition, Wiley International Edition
2. Venkateshmurthy, "*Introduction to Unix and Shell Programming*", Pearson Education (India)
3. Stallings W, "*Operating Systems*",7th Edition, Pearson Education
4. Bach M J, "*The Design of Unix Operating System*", Prentice Hall India
5. Flynn I. M, "*Understanding Operating Systems*", Cengage India Publication
6. Tanenbaum A.S., "*Modern Operating Systems*",4th Edition, Pearson Education India
7. Eric Foster, John Welch, Micah Anderson, "*Beginning Shell Scripting*", Wrox Publication
8. Sibsankar Haldar, Alex Aravind, "*Operating Systems*", Pearson Education

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Apply practical knowledge of Linux System and shell scripting
- Harness the facilities provided by Operating System in application development
- Optimize the programs for specific Operating System

***** ***** *****